

1 Features of a time series

Univariate: A time series that consists of a single variable recorded sequentially over time (e.g., bike sale over time from a single store) Multivariate: Consists of multiple variables recorded sequentially over time (e.g., bike sale and profit over time from a single store) Hierarchical: Consists of multiple variables from multiple observations, nested within the larger group categories, recorded sequentially over time (e.g., bike sale and profit from multiple stores over time)

1.1 Different tasks in time series analysis

Prediction/forecasting (supervised) Example: Predicting the number of tourists in the upcoming quarters, Approach: Statistical models (exponential smoothing, ARIMA, etc..) or ML/DL models (CNN,RNN,LightGBM, etc...)

Clustering/Anomaly detection (unsupervised) Example 1: Detect earthquakes, Approach: Start with some dimensionality reduction/transformation, compute similarity measures (euclidean, DTW), and apply unsupervised learning models (k-means, DBSCAN, etc.)

Factors that affect forecasting

- How much (historical) data we have
- How far into the future do we forecast
- How similar historical patterns are compared to the future
- How 'stable' the environment is
- How much domain knowledge/information you possess

1.2 Working with time series - Visualizing

The first step in any time series analysis should be a time plot! **Temporal dependence** What makes time series different to data we've seen previously is that they include temporal dependence - observations close in time tend to be correlated. That means that the value of the series at time t often depends in some way on past values. We can quantify this dependence by looking at the correlation of a time series with "lagged" values of itself. We call this **autocorrelation**.

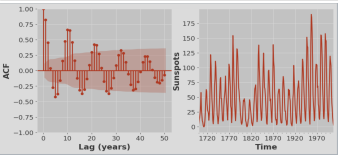
```
sun["sunspots (lag=1)"] = sun["sunspots"].shift(1)
```

Correlogram Simpler plot to visualise all this information. Plots the autocorrelation function (ACF) on the y-axis and lags on the x-axis. The formula used for standard error depends upon the situation. If ACF for residuals diagnostics as part of the ARIMA routine, the

standard errors are determined assuming the residuals are white noise..

If ACF used as part of the raw data as a model selection tool, then we should use the Bartlett's formula. 'bartlett_confint=True'

```
fig, (ax1,ax2) = plt.subplots(nrows=1, ncols=2, figsize=(11, 5))
plot_acf(sun["sunspots"], lags=50, title=None,ax=ax1)
ax1.set_xlabel("Lag (years)")
ax1.set_ylabel('ACF')
sun.sunspots.plot(line(xlabel="Time", ylabel="Sunspots",
                      legend=False,ax=ax2);
```



1.3 Time Series Patterns